1 MDI Server Configuration

1.1 Overview

In order to collect data online in the product groups FEP, WEP, EMU and PMV, you have to do the following:
- install and configure one or several MDI servers
- configure MDI in the MOC

The training course EAT-MDI provides further information on the MDI servers you should use in each specific case. In general, you can choose from the following MDI servers:

- MDI Steinwald
- MDI IBRit
- MDI Seriell (serial)
- MDI Liste Seriell (serial list)
- MDI Messwertliste (measurement list)
- MDI Messwertdatei (measurement file)

Contact MPDV to request the MDI servers.

You can install each MDI server separately by starting the file "setup.exe". You need local administration rights to install the servers.

The computer where you want to install the MDI server must have a fixed IP address (not DHCP) or you must configure the MDI server as local host.

You must stop all running MDI servers prior to the installation.

You have to exit the MDI server via the menu item "File --> Exit" to make sure all settings will be saved.

Note the following if you collect measured values of several characteristics:

You must not access the same MDI from several AIP inspection stations at the same time. Consequently, several AIP inspection stations cannot export data via the same MDI.

Enable the MDI queue mode, if several AIP inspection stations access the same MDI. The MDI queue mode is available as of service pack 14. Enable/activate the MDI queue mode manually. Section "MDI queue mode" of this procedure document provides further information on this.

You have to activate the licenses AIP-MDI and SIS-IMM in order to use online data collection.

After installing and starting the MDI server, the MDI server is displayed as an icon in the status area of the taskbar. A hammer and a red/green LED represent the server. A red LED indicates that the MDI server is not active. If the LED is green, the MDI server is active.

A tooltip appears, if you mouse over the icon. The tooltip shows the name of the MDI server and a list of the channels.

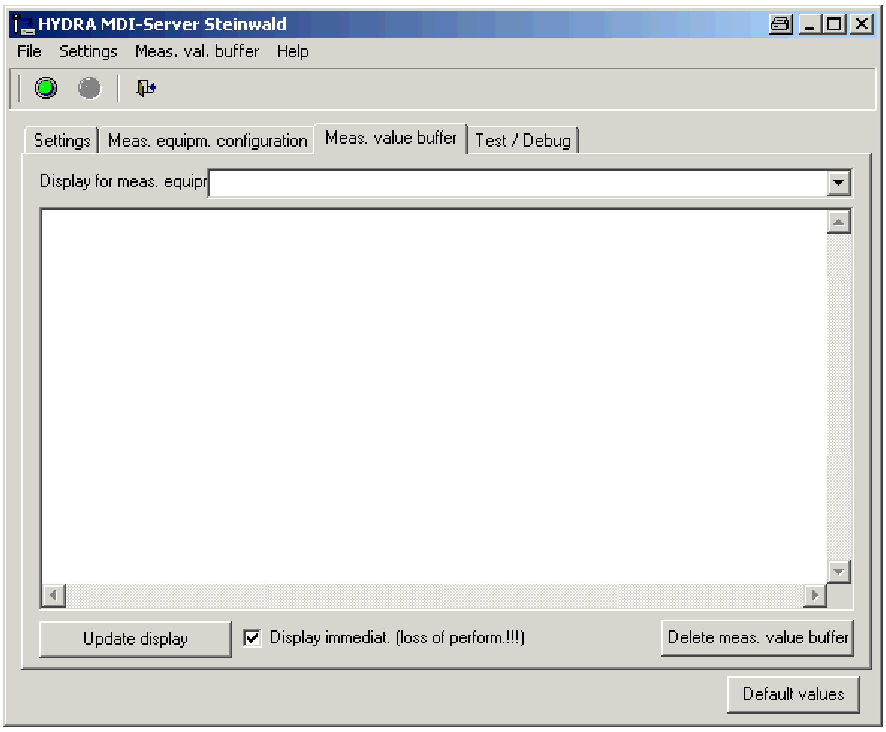
A single left-click opens the MDI server dialog. Right-click to open the context menu. You can also use the context menu to open the MDI server dialog. You can also use the context menu to enable, disable or close the MDI server.

Disable the MDI server if you want to change the general settings or the settings of the file structure. The LED of the MDI server icon will then appear red in the status area of the taskbar.

Use the menu item *File* to enable, disable or close the MDI server. Also use this menu item to configure the language.

Use the menu item *Settings* to restore the default values of the MDI server. You can also import and export configuration settings.

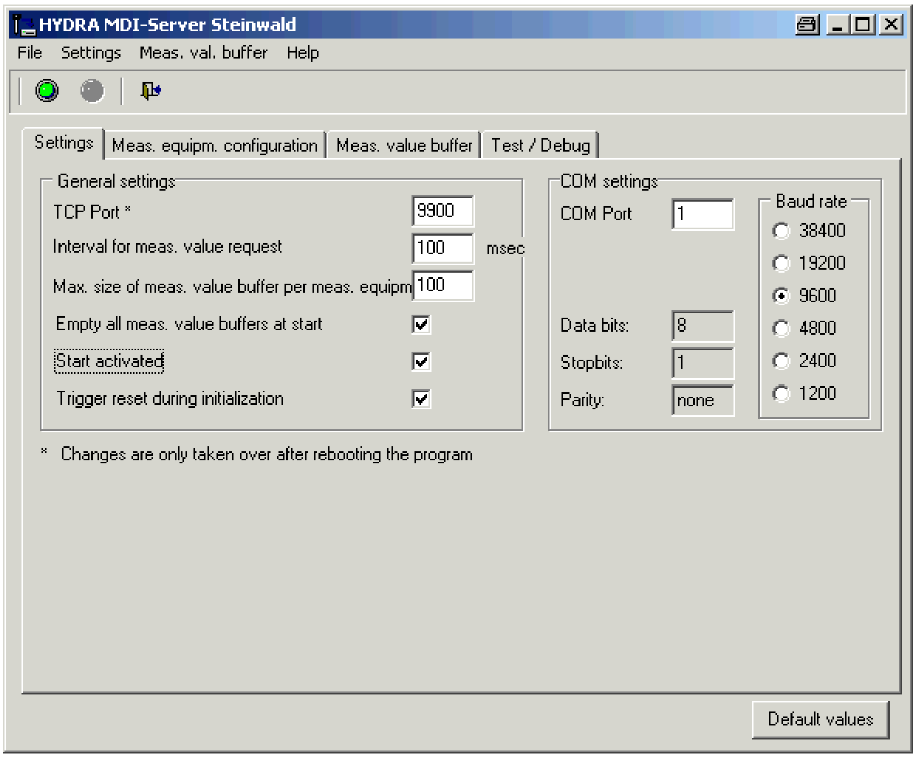
Use the menu item *Measurement buffer* to delete the measurement buffers of all channels.



The *Measurement buffer* tab shows the data deriving from the measuring equipment. To do so, select a characteristic/channel in the selection list "Characteristics display" and click the *Refresh* button. Then the measurement buffer is displayed for this characteristic/channel. If the *Display immediately* option is set, the displayed data is constantly updated/refreshed. This might lead to performance losses. Select the button "Delete measurement buffer" to delete the measurement buffer for the selected characteristic/channel. This process does not delete the measurement buffers of the other characteristics/channels.

The *Test/debug* tab shows information on the TCP interface and internal MDI server processing. You can use this information for troubleshooting or test purposes. Usually, this information is not relevant for normal operation.

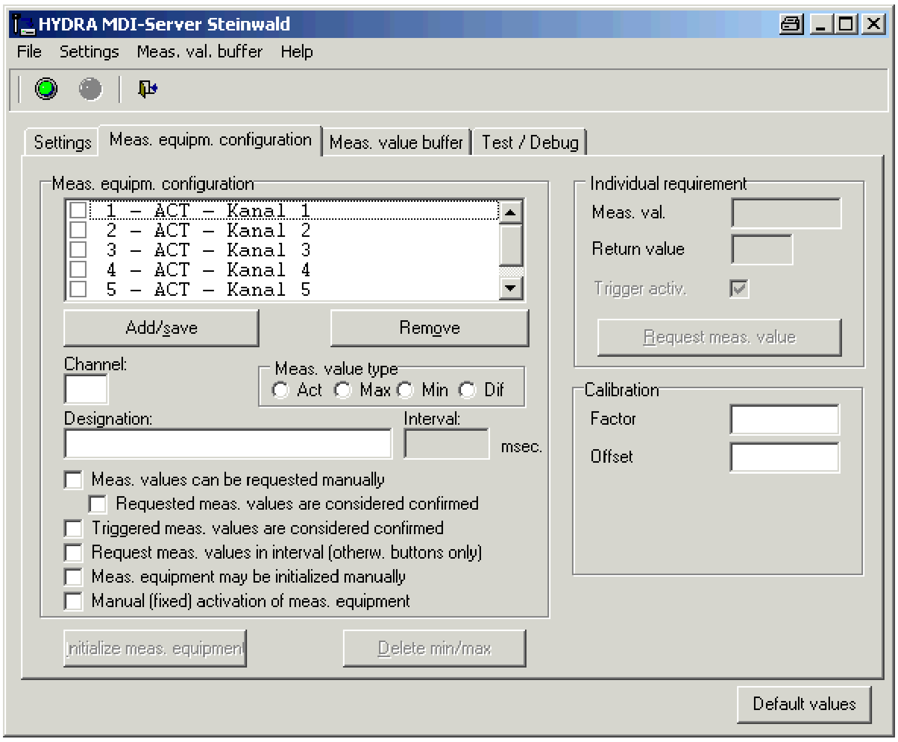
1.2 MDI Steinwald



Use the *Settings* tab to define the TCP port, the interval for the measured value query, the maximum measured value buffer size per measuring equipment/characteristic and the COM settings. In addition, you can specify if:

- all measured value buffers are emptied when the server is started,
- the MDI server is activated on starting
- a reset is triggered on starting. Any changes to the settings are saved locally when the MDI server is closed (not when it is deactivated).

The COM port settings depend on the connected Steinwald box and can be found in the corresponding manual. If the Steinwald box is connected via USB, enter the COM parameters of the corresponding USB serial driver. Refer to the manufacturer's documentation for further information.
The entered TCP port must match the configuration of the communication port (MOC Quality Management --> Master data --> MDI Configuration --> MDI Configuration).



You can create and/or edit the characteristics/channels in the section *Measurement equipment configuration.* If you want to create a new characteristic/channel, fill out the input fields *Channel, Measured value type* and *Designation/Name* and configure the required settings. Finally, click the *Add/Save* button.

Measured value type:

Act: Current measured value

Max: Maximum measured value

Min: Minimum measured value

Dif: Difference between maximum and minimum measured value

Measured values can be requested manually: If the checkbox is enabled, the MDI client can request a measured value from the measuring equipment.

Requested measured values are considered confirmed: You can only select this option, if you also checked the option “Measured values can be requested manually”. In this case, a measured value requested by the MDI client is considered confirmed. Measured values deriving from the measuring equipment and not requested by the MDI client are considered unconfirmed.

Triggered measured values are considered confirmed: Select this option to confirm all measured values that are triggered by foot switches or device buttons.

Request measured values at intervals (otherwise buttons only): Set this option to request data from the connected measuring equipment at regular intervals (polling). Enter the interval in milliseconds in the relevant input field. The default value is 100 milliseconds.

Measuring equipment may be initialized manually: Set this option to allow the MDI client to initialize the measuring equipment of this channel.

Manual (fixed) activation of measuring equipment: Set this option to activate or deactivate the measuring equipment/channel permanently, depending on whether the option is set or not for the channel itself. Consequently, the MDI server ignores any request sent by the MDI client to the MDI server to enable or disable the connected measuring equipment. The MDI client is sent a message indicating that this functionality is not supported. If the option is set, the channel status (active/inactive) is saved when the MDI server is closed and restored when it is started the next time.

Please note: Changes to the characteristics/channels only take effect after you clicked the *Add/save* button!

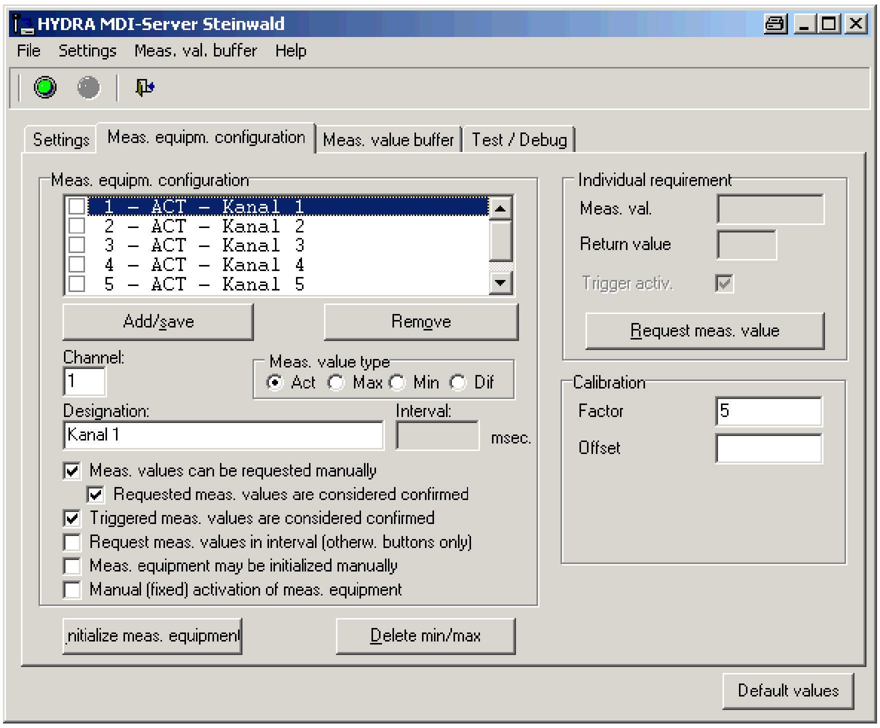
The list then shows the measuring equipment (characteristic) with the channel, the type and the designation/name. A checkbox is also displayed. You can use this checkbox to activate or deactivate the characteristic/channel.

Calibration:

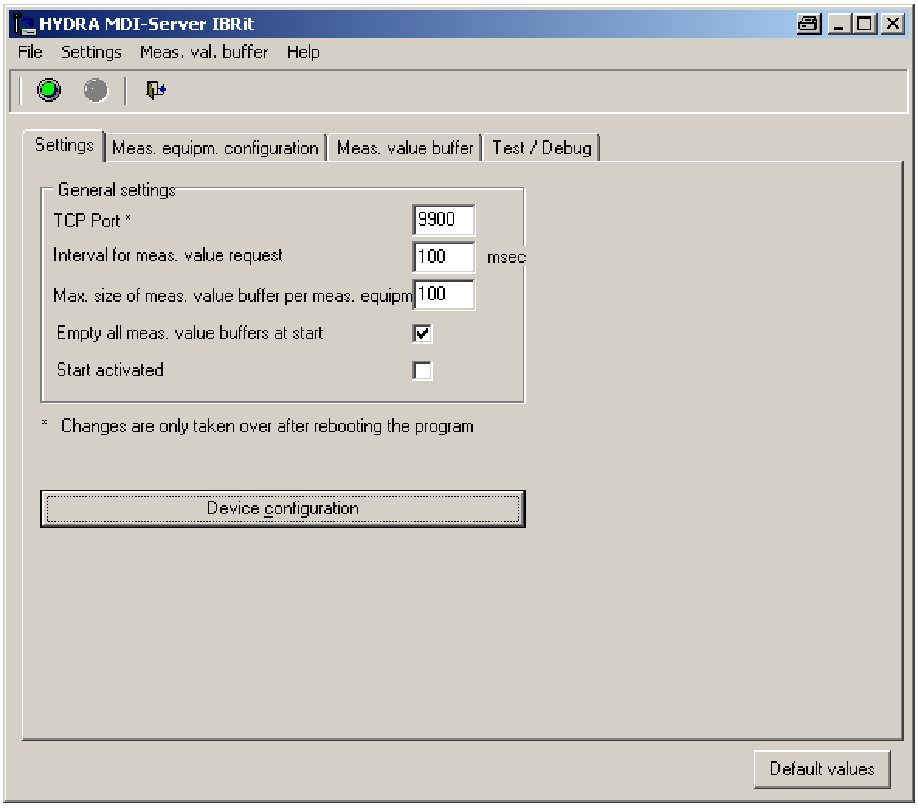
You can enter an integer or decimal value in the *Factor* input field. You can multiply the incoming measured value of the channel with this value. You can also enter an integer or decimal value in the *Offset* input field. You can add to or subtract this value from the incoming measured value of the channel.

You can only make entries in the fields *Factor* and *Offset* if the MDI server is not enabled. Click the *Add / Save* button to store the entered values for the channel.

Please note: Thousands separators are not supported. Commas or points are required as decimal separators.


HYDRA MDI-Server Steinwald
File
Settings
Meas. val. buffer
Help
Settings
Meas. equipm. configuration
Meas. value buffer
Test / Debug
Meas. equipm. configuration
1 – ACT – Kanal 1
2 – ACT – Kanal 2
3 – ACT – Kanal 3
4 – ACT – Kanal 4
5 – ACT – Kanal 5
Add/save
Remove
Channel:
1
Meas. value type
Act
Max
Min
Dif
Designation:
Kanal 1
Interval:
msec.
Meas. values can be requested manually
Requested meas. values are considered confirmed
Triggered meas. values are considered confirmed
Request meas. values in interval (otherw. buttons only)
Meas. equipment may be initialized manually
Manual (fixed) activation of meas. equipment
nitialize meas. equipment
Delete min/max
Individual requirement
Meas. val.
Return value
Trigger activ.
Request meas. value
Calibration
Factor
5
Offset
Default values

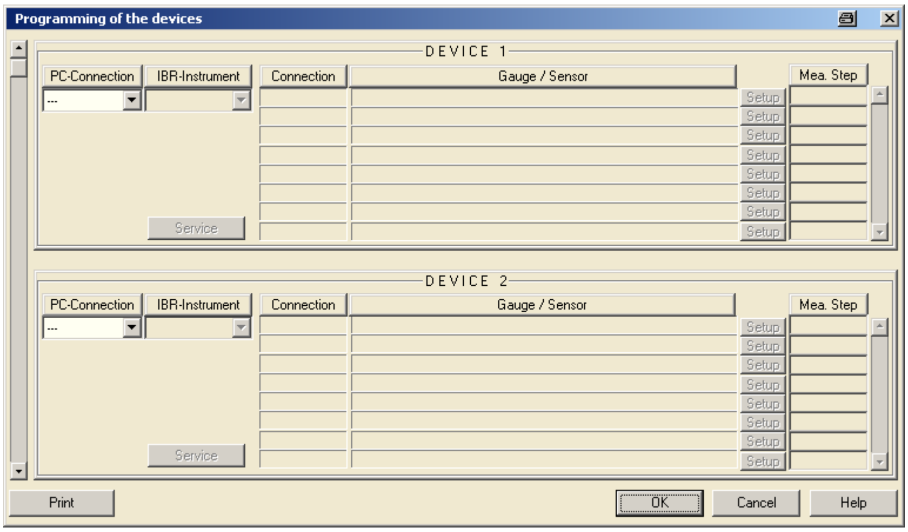
1.3 MDI IBRit

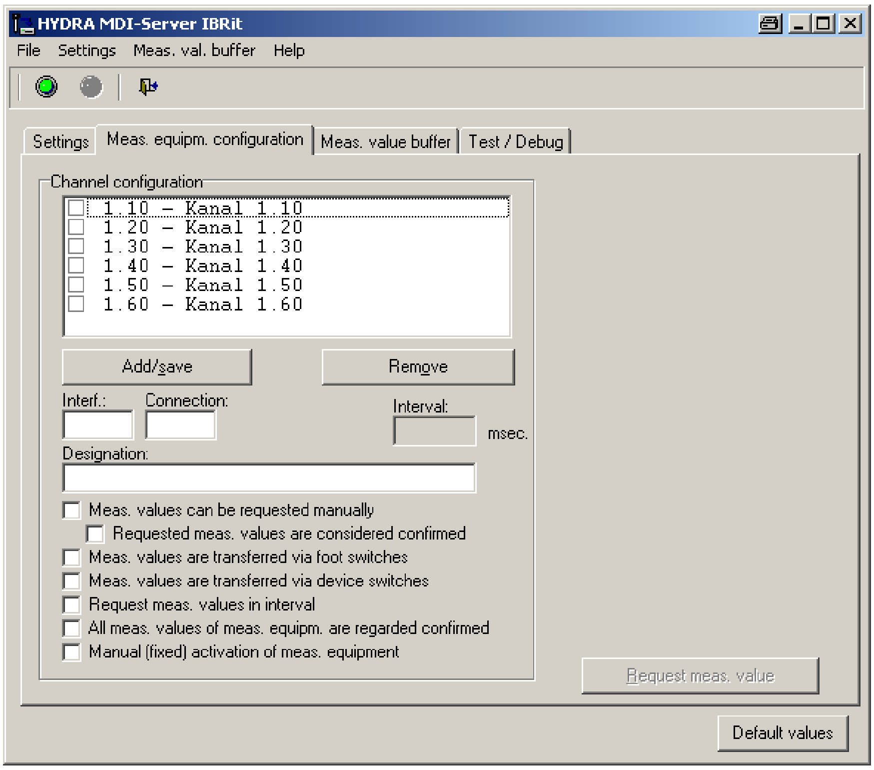

HYDRA MDI-Server IBRit
File
Settings
Meas. val. buffer
Help
Settings
Meas. equipm. configuration
Meas. value buffer
Test / Debug
General settings
TCP Port *
9900
Interval for meas. value request
100
msec
Max. size of meas. value buffer per meas. equipm
100
Empty all meas. value buffers at start
Start activated
* Changes are only taken over after rebooting the program
Device configuration
Default values

Use the *Settings* tab to specify the TCP port, the interval for the measured value query and the maximum size of the measured value buffer per measuring equipment/characteristic. You can also specify if the MDI server is activated upon starting and if all measured value buffers are emptied. Any changes to the settings are saved locally when the MDI server is closed (not when it is deactivated).

The entered TCP port must match the configuration of the communication port (MOC Quality Management --> Master data --> MDI Configuration --> MDI Configuration).

Click the button *Device configuration* to call the IBRit configuration program and to configure the interface and connected devices. The IBRit manual provides a detailed description.





You can create and/or edit the characteristics/channels in the section *Measurement equipment configuration.* If you want to create a new characteristic, fill out the input fields *Interface, Connection* and *Designation/Name* and configure the required settings. Then click the *New/save* button.

Measured values can be requested manually: If the checkbox is enabled, the MDI client can request a measured value from the measuring equipment.

Requested measured values are considered confirmed: You can only select this option, if you also checked the option "Measured values can be requested manually". In this case, a measured value requested by the MDI client is considered confirmed. Measured values deriving from the measuring equipment and not requested by the MDI client are considered unconfirmed.

Measured values are transferred via foot switches: If you check this option, the current measured value will be transferred once you have used the foot switch.

Measured values are transferred via device switches: If you select this option, the current measured value will be transferred once you have used the data transfer switch of the measuring equipment, if available.

Request measured values at intervals: Set this option to request data from the connected measuring equipment at regular intervals (polling). Enter the interval in milliseconds in the relevant input field. The default value is 100 milliseconds.

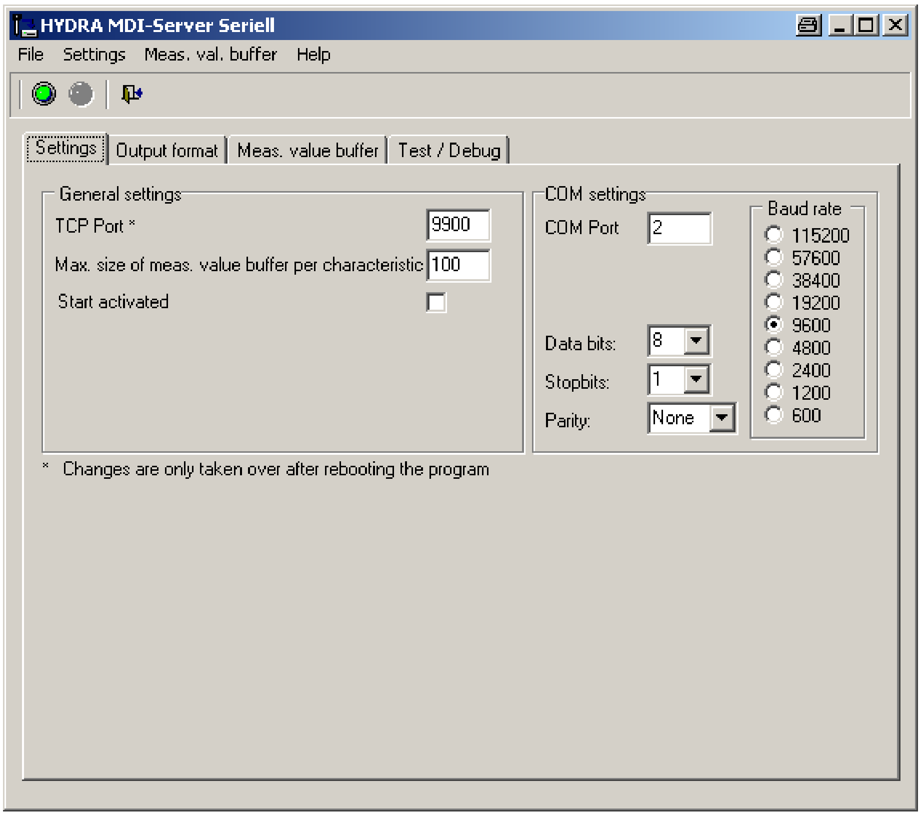
All measured values of the measuring equipment are considered confirmed: All measured values deriving from the connected measuring equipment are considered confirmed.

Manual (fixed) activation of measuring equipment: Set this option to activate or deactivate the measuring equipment/channel permanently, depending on whether the option is set or not for the channel itself. Consequently, the MDI server ignores any request sent by the MDI client to the MDI server to enable or disable the connected measuring equipment. The MDI client is sent a message indicating that this functionality is not supported. If the option is set, the channel status (active/inactive) is saved when the MDI server is closed and restored when it is started the next time. Please note that the channel status (active/inactive) is only saved for such channels where the option for the manual (fixed) activation is enabled.

Please note: Changes to the characteristics/channels only take effect after you clicked the *New/save* button!

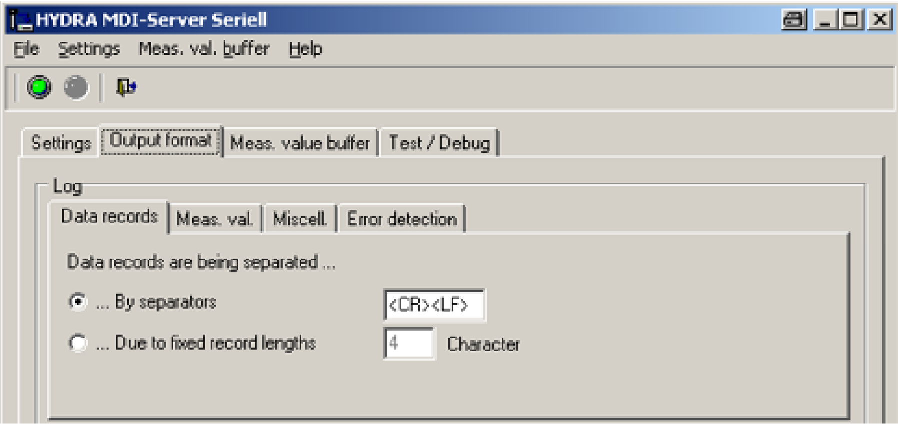
The list then shows the measuring equipment (characteristic) with the interface, connection and designation/name. A checkbox is also displayed. You can use this checkbox to activate or deactivate the characteristic/channel.

1.4 MDI Seriell (serial)



Use the *Settings* tab to specify the TCP port, the maximum measured value buffer size per measuring equipment/characteristic and the COM settings. You can also specify if the MDI server should be activated immediately upon starting. Any changes to the settings are saved locally when the MDI server is closed (not when it is deactivated).

The COM port settings depend on the measuring equipment you connect. The relevant measuring equipment manuals provide further details on these settings. The entered TCP port must match the configuration of the communication port (MOC Quality Management --> Master data --> MDI Configuration --> MDI Configuration).



Use the *Output format* tab to specify the structure of data records coming from the measuring equipment. You can also find this information in the measuring equipment manual.

If you want to enter special characters and/or control characters, use the synonyms listed below:

<SOH> corresponds to ASCII character 1

<STX> corresponds to ASCII character 2

<ETX> corresponds to ASCII character 3

<EOT> corresponds to ASCII character 4

<ACK> corresponds to ASCII character 6

<TAB> corresponds to ASCII character 9

<LF> corresponds to ASCII character 10

<FF> corresponds to ASCII character 12

<CR> corresponds to ASCII character 13

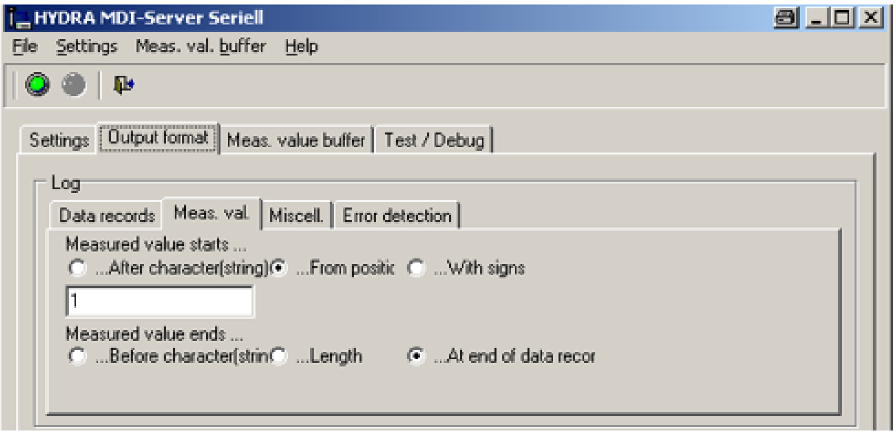
<NACK> corresponds to ASCII character 21

<EOF> corresponds to ASCII character 26

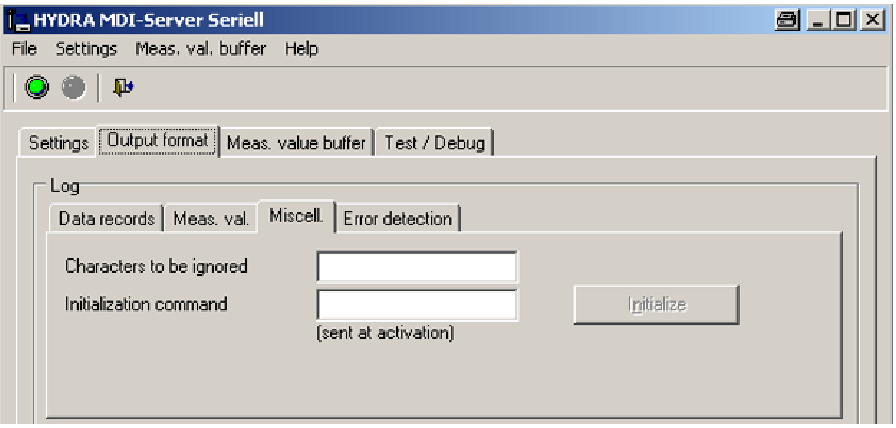
<SPACE> corresponds to ASCII character 32

Example: Enter <CR><LF> in the corresponding input field, if you want to separate the data records sent by the measuring equipment using carriage return and line feed.

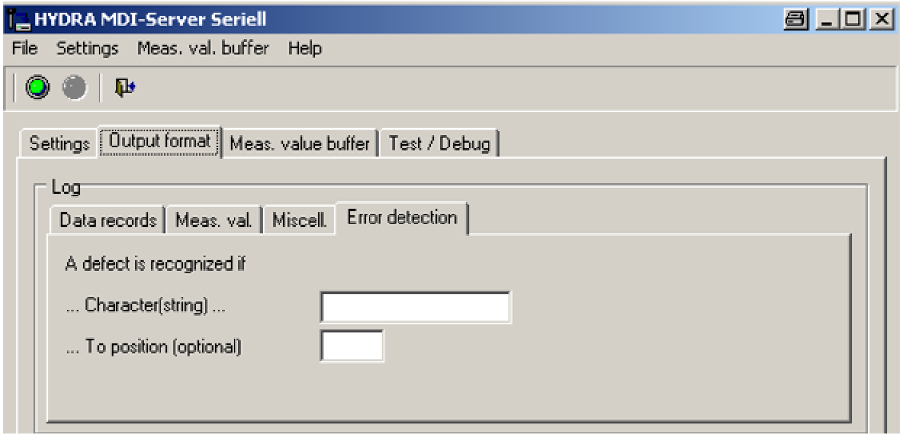
You can separate data records using separators or by specifying a fixed record length.



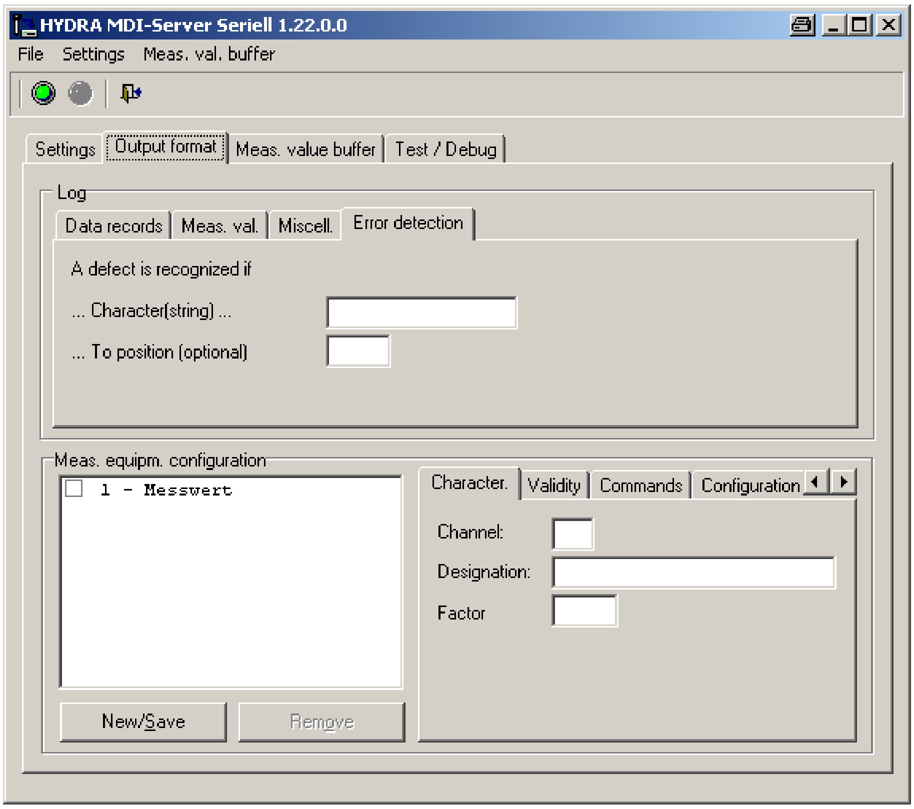
Use the *Measured value* tab to specify how the measured value is identified in a data record. You can also find this information in the measuring equipment manual.



Specify the characters to be ignored and the initialization command of the measuring equipment (if available) in the *Miscellaneous* tab. Characters to be ignored will be removed from the data record before the data record will actually be processed. The initialization command is sent to the measuring equipment when the MDI server is enabled or the respective button is clicked.



Use the *Error detection* tab to specify character strings that identify a data record as faulty. If the measuring equipment supports this function, you can find these character strings in the manual about the measuring equipment.



You can create and/or edit the characteristics/channels in the section *Measuring equipment configuration*. If you want to create a new characteristic/channel, fill out the input fields *Channel* and *Name/Designation.* You also have to make your required entries in the tabs *Validity, Commands* and *Configuration*. Then click the *New/Save* button.

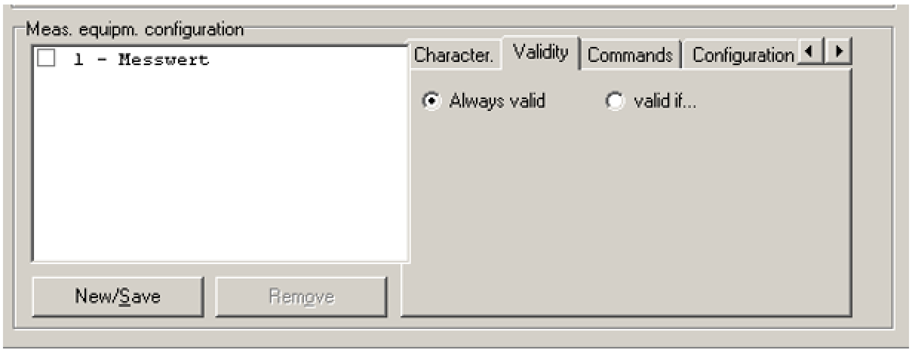
The input field *Factor* only appears if the configuration file *mdiseriell.ini* is configured accordingly. If the field should be displayed, the segment [General] must include the entry FactorUsed=1. The measured value is multiplied by the value included in the *Factor* field. The factor value can be a positive, negative, integer or floating-point number. Factor 1 will be used, if no factor is indicated for a characteristic. You can enter a comma as decimal separator. However, a decimal point will be stored and used.

Please note: Changes to the characteristics/channels only take effect after you clicked the *New/save* button!

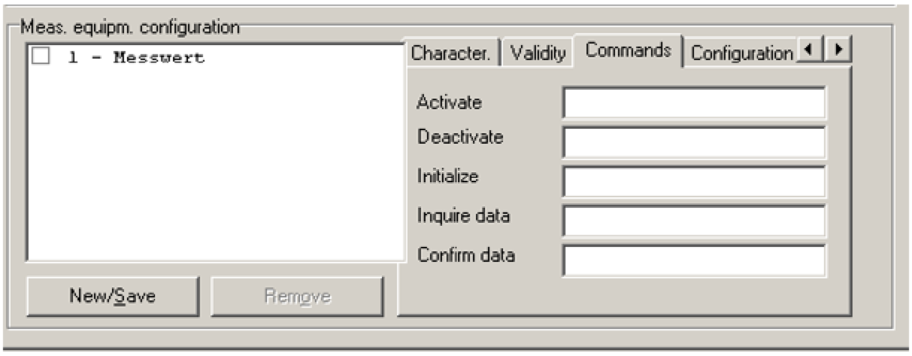
The list now shows the measuring equipment (characteristic) including the channel number and the designation/name. A checkbox is also displayed. You can use this checkbox to activate or deactivate the characteristic/channel.

Please note: Activating and/or deactivating a piece of measuring equipment/channel is not saved unless you configured the manual (fixed) activation for this measuring equipment/channel.

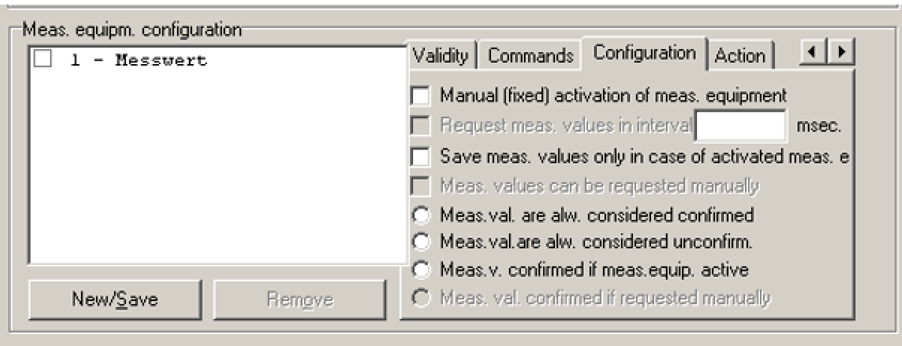
Also see the section about the *Configuration* tab.



Use the *Validity* tab to specify if a measured value that is sent by the measuring equipment is always valid or if the data record must include a specific character string that identifies a measured value as valid.



Use the *Commands* tab to specify different commands for each characteristic/channel, provided that these commands are supported by the connected measuring equipment. If you enter a command in the *Activate* field, for example, this command will be sent to the connected measuring equipment as soon as the MDI client sends a request to the MDI server.



Use the *Configuration* tab to configure each characteristic/channel separately.

Manual (fixed) activation of measuring equipment: Set this option to activate or deactivate the measuring equipment/channel permanently, depending on whether the option is set or not for the channel itself. Consequently, the MDI server ignores any request sent by the MDI client to the MDI server to enable or disable the connected measuring equipment. The MDI client is sent a message indicating that this functionality is not supported. If the option is set, the channel status (active/inactive) is saved when the MDI server is closed and restored when it is started the next time. Please note that the channel status (active/inactive) is only saved for such channels where the option for the manual (fixed) activation is enabled.

Request measured values at intervals: You can only check this option, if you entered a command for this characteristic in the *Request data* input field of the *Commands* tab. If this is the case and the option is checked, then data is requested at regular intervals from the connected measuring equipment (polling). To do so, the command entered in the *Request data* input field is sent to the connected measuring equipment at regular intervals. Enter the interval in milliseconds in the relevant input field. The default value is 100 milliseconds.

Save measured values only for activated measuring equipment: If this option is checked, measured values deriving from measuring equipment will be ignored if the channel is not active.

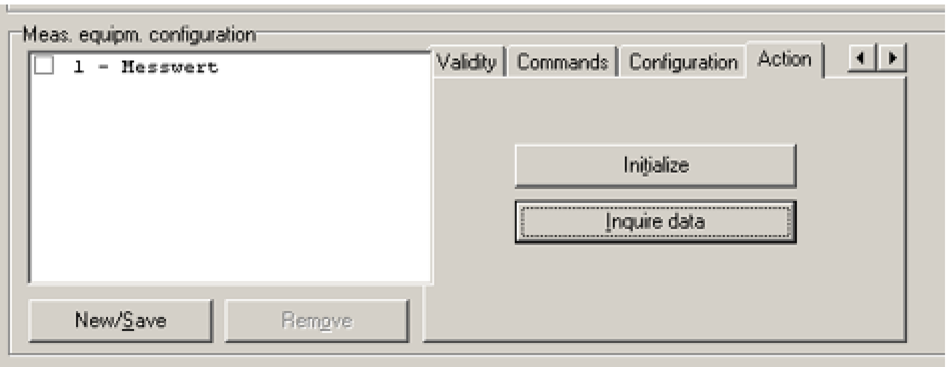
Measured values can be requested manually: You can only check this option, if you entered a command for this channel/characteristic in the *Request data* input field of the *Commands* tab. If this is the case and the option checked, the MDI client can request a measured value. The MDI server sends the command specified in the *Request data* input field to the connected measuring equipment.

Measured values are always considered confirmed: All measured values deriving from the connected measuring equipment are considered confirmed for this channel/characteristic.

Measured values are always considered unconfirmed: All measured values deriving from the connected measuring equipment are considered unconfirmed for this channel/characteristic.

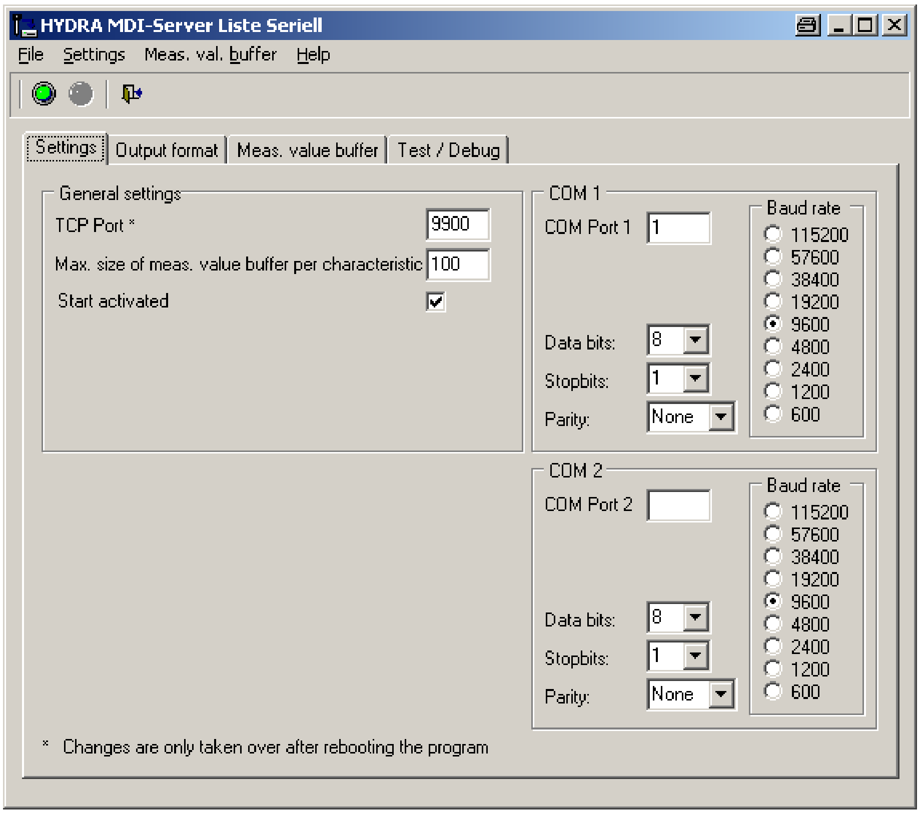
Measured value confirmed if measuring equipment is active: All measured values deriving from the connected measuring equipment are considered confirmed for this channel/characteristic, provided that the channel/characteristic is active. If the channel/characteristic is inactive, the measured values are considered unconfirmed.

Measured value confirmed if requested manually: You can only select this option, if you also checked the option *Measured values can be requested manually.* In this case, a measured value requested by the MDI client is considered confirmed. Measured values deriving from the measuring equipment and not requested by the MDI client are considered unconfirmed.



Use the *Action* tab to initialize the channel/characteristic or to request data. To do so, you have to make the corresponding entries in the input fields *Initialize* and/or *Request data* of the *Commands* tab.

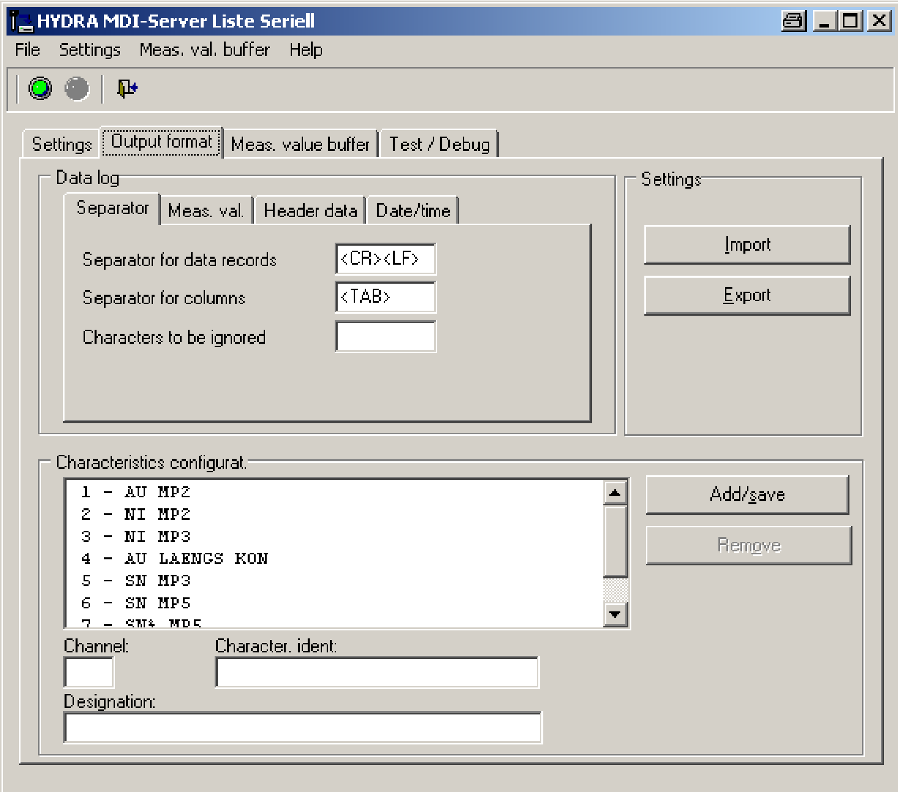
1.5 MDI Liste Seriell (serial list)



Use the *Settings* tab to define the TCP port, the maximum measured value buffer size per measuring equipment/characteristic and the COM settings. 2 COM ports are supported. Leave the fields *COM port 1* or *COM port 2* empty if you only want to use one COM port.

You can also specify if the MDI server should be activated immediately upon starting. Any changes to the settings are saved locally when the MDI server is closed (not when it is deactivated).

The COM port settings depend on the measuring equipment you connect. The relevant measuring equipment manuals provide further details on these settings. The entered TCP port must match the configuration of the communication port (MOC Quality Management --> Master data --> MDI Configuration --> MDI Configuration).



You can create and/or edit the characteristics/channels in the section Characteristic configuration. If you want to create a new characteristic/channel, fill out the input fields Channel, Characteristic ID and Designation (name). Then click the *Add/save* button.

Please note: Changes to the characteristics/channels only take effect after you clicked the *Add/save* button!

The list now shows the measuring equipment (characteristic) including the channel number and the designation/name.

Use the tabs *Output format / Separators* to specify the structure of data records deriving from the measuring equipment. You can also find this information in the measuring equipment manual.

Characters to be ignored will be removed from the data record before the data record will actually be processed.

If you want to enter special characters and/or control characters, use the synonyms listed below:

<SOH> corresponds to ASCII character 1

<STX> corresponds to ASCII character 2

<ETX> corresponds to ASCII character 3

<EOT> corresponds to ASCII character 4

<ACK> corresponds to ASCII character 6

<TAB> corresponds to ASCII character 9

<LF> corresponds to ASCII character 10

<FF> corresponds to ASCII character 12

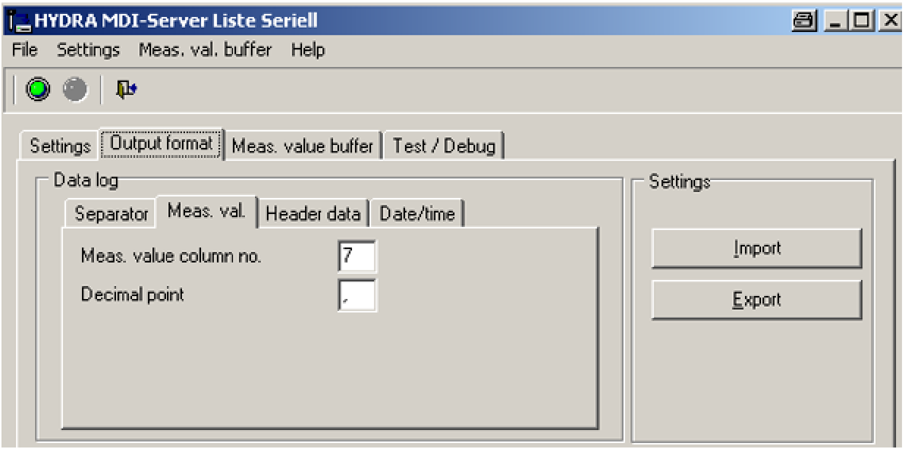
<CR> corresponds to ASCII character 13

<NACK> corresponds to ASCII character 21

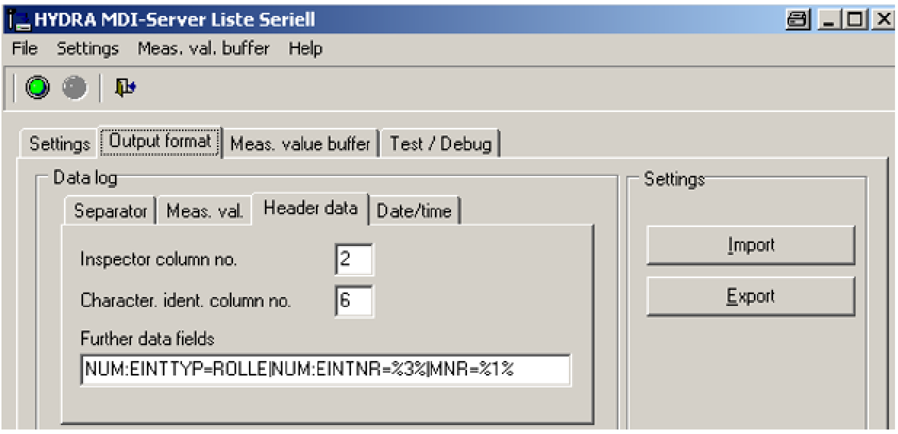
<EOF> corresponds to ASCII character 26

<SPACE> corresponds to ASCII character 32

Example: Enter <CR><LF> in the corresponding input field, if you want to separate the data records sent by the measuring equipment using carriage return and line feed.



Use the *Measured value* tab to specify the column of the data record that includes the measured value and the character that is used as the decimal separator. You can also find this information in the measuring equipment manual.



Use the *Header data* tab to specify the column number of the *Inspector* and the *Characteristic ID.*

You can also specify further data fields. Use the pipe character "|" to separate these data fields. Use percent signs to indicate the column number of the data field.

Example: MNR=%1%

The machine number (MNR) is entered in the first column of the data record sent by the measuring equipment.

The terminal supports the following data fields to filter the collected data:

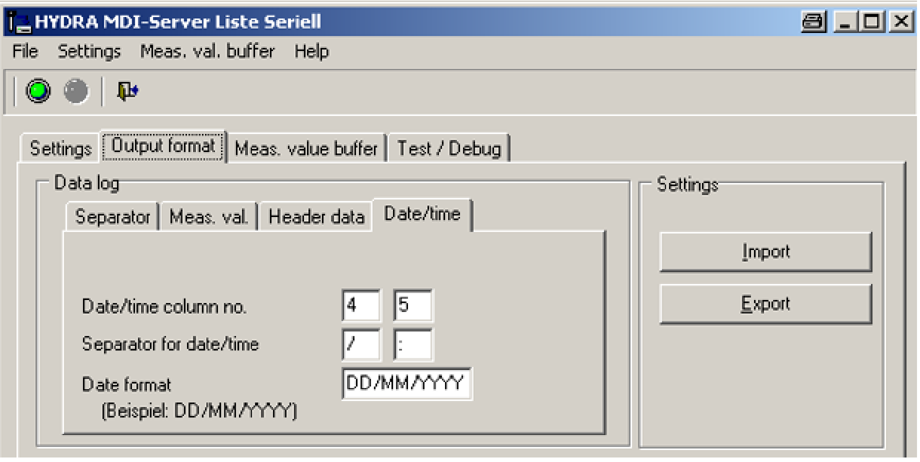
CNR	batch number
ANR	order number
ATK	article number
NUM:EINTTYP	number type for measurement recording of numbers, e. g. PPUNKT for inspection point
NUM:EINTNR	number, e. g. inspection point number 000001

Example of data collection via the terminal:

- The data field *CNR* is configured and the measurement buffer of a channel includes measured values with different batch numbers.

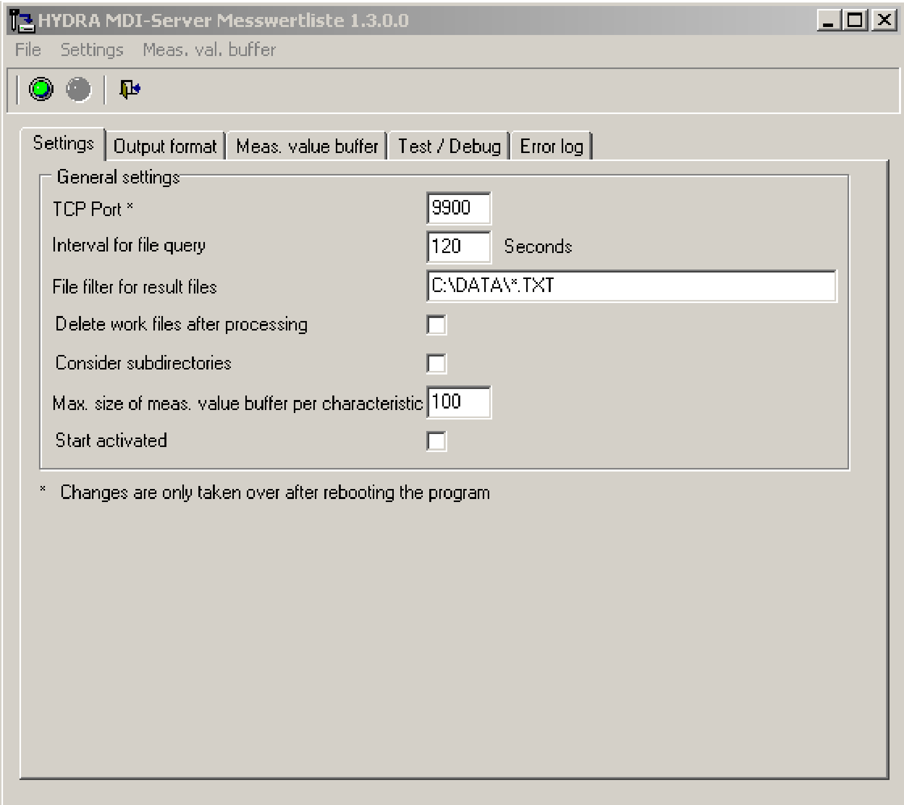
- If you open measurement data collection and request measured values from the MDI server, only those measured values are collected and saved whose batch number of the MDI measurement buffer match that of the inspection requirement.

- If the inspection requirement does not include a batch number, the filter is not enabled and all measured values are collected from the measurement buffer.



Use the *Date/time* tab to specify the column numbers and/or separators for the date, time and the date format.

1.6 MDI Messwertliste (measurement list)

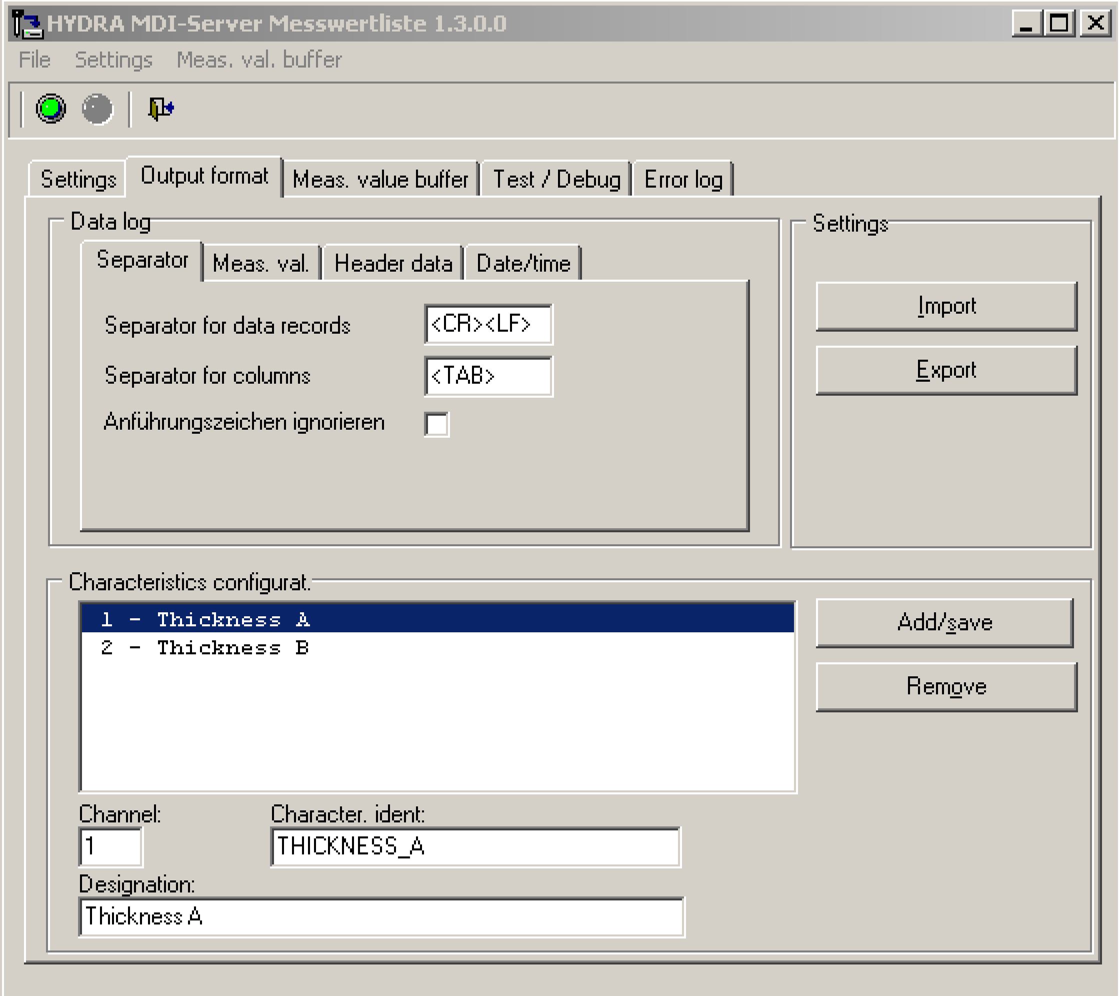


This MDI has been designed to collect measured values from all characteristics. You should only use this MDI for this purpose. If you use the MDI for characteristics and you select the characteristic in the AIP inspection data collection, the system automatically requests the measured values, enters and saves these values in the measurement field. Please note that the configurable additional data fields are not used to filter the collected data. In this context, AIP inspection data collection always requests and collects all data.

Use the *Settings* tab to define the TCP port, the interval for the file query and the file filter for result files. You can also specify if work files are deleted after processing, if subdirectories are integrated and if the MDI server is to be activated immediately upon starting. You can also specify the maximum size of the measurement buffer for each piece of measuring equipment/characteristic.

Any changes to the settings are saved locally when the MDI server is closed (not when it is deactivated).

The COM port settings depend on the measuring equipment you connect. The relevant measuring equipment manuals provide further details on these settings. The entered TCP port must match the configuration of the communication port (MOC Quality Management --> Master data --> MDI Configuration --> MDI Configuration).



You can create and/or edit the characteristics/channels in the section *Characteristic configuration*. If you want to create a new characteristic/channel, fill out the input fields *Channel, Characteristic ID* and *Designation (name).* Then click the *Add/save* button.

Please note: Changes to the characteristics/channels only take effect after you clicked the *Add/save* button!

The list now shows the measuring equipment (characteristic) including the channel number and the designation/name.

Use the tabs *Output format / Separators* to specify the structure of data records deriving from the measuring equipment. You can also find this information in the measuring equipment manual.

If you want to enter special characters and/or control characters, use the synonyms listed below:

<SOH> corresponds to ASCII character 1

<STX> corresponds to ASCII character 2

<ETX> corresponds to ASCII character 3

<EOT> corresponds to ASCII character 4

<ACK> corresponds to ASCII character 6

<TAB> corresponds to ASCII character 9

<LF> corresponds to ASCII character 10

<FF> corresponds to ASCII character 12

<CR> corresponds to ASCII character 13

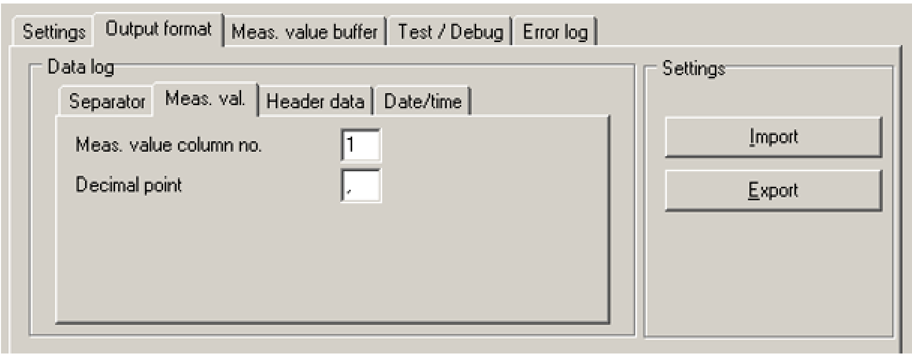
<NACK> corresponds to ASCII character 21

<EOF> corresponds to ASCII character 26

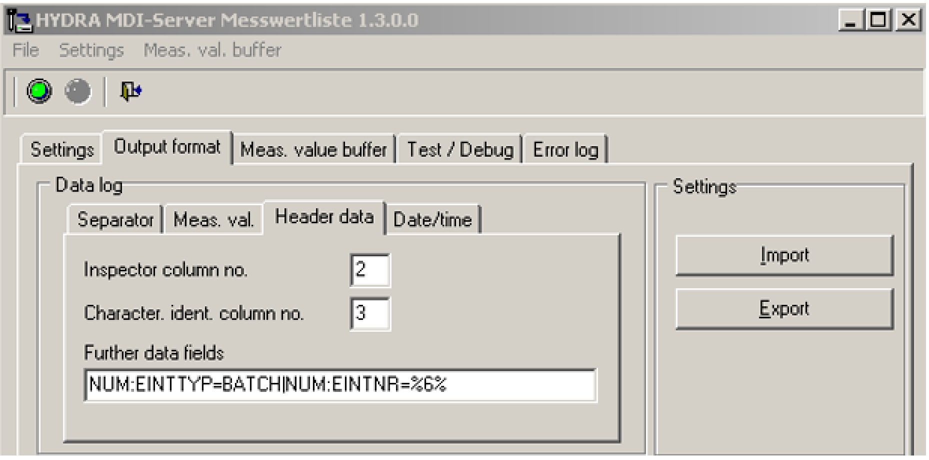
<SPACE> corresponds to ASCII character 32

Example: Enter <CR><LF> in the corresponding input field, if you want to separate the data records sent by the measuring equipment using carriage return and line feed.

Use the checkbox *Ignore quotes* to specify whether or not double quotes (ASCII #34) as they are used, for example in ".csv" files to restrict strings are integrated while processing the file. If they are not integrated, the characteristic ID is stated without quotation marks.



Use the *Measured value* tab to specify the column of the data record that includes the measured value and the character that is used as the decimal separator. You can also find this information in the measuring equipment manual.



Use the *Header data* tab to specify the column number of the *Inspector* and the *Characteristic ID.*

You can also specify further data fields. Use the pipe character "|" to separate these data fields. Use percent signs to indicate the column number of the data field.

Example: MNR=%2%

The machine number (MNR) is entered in the second column of the data record sent by the measuring equipment.

The terminal supports the following data fields to filter the collected data:

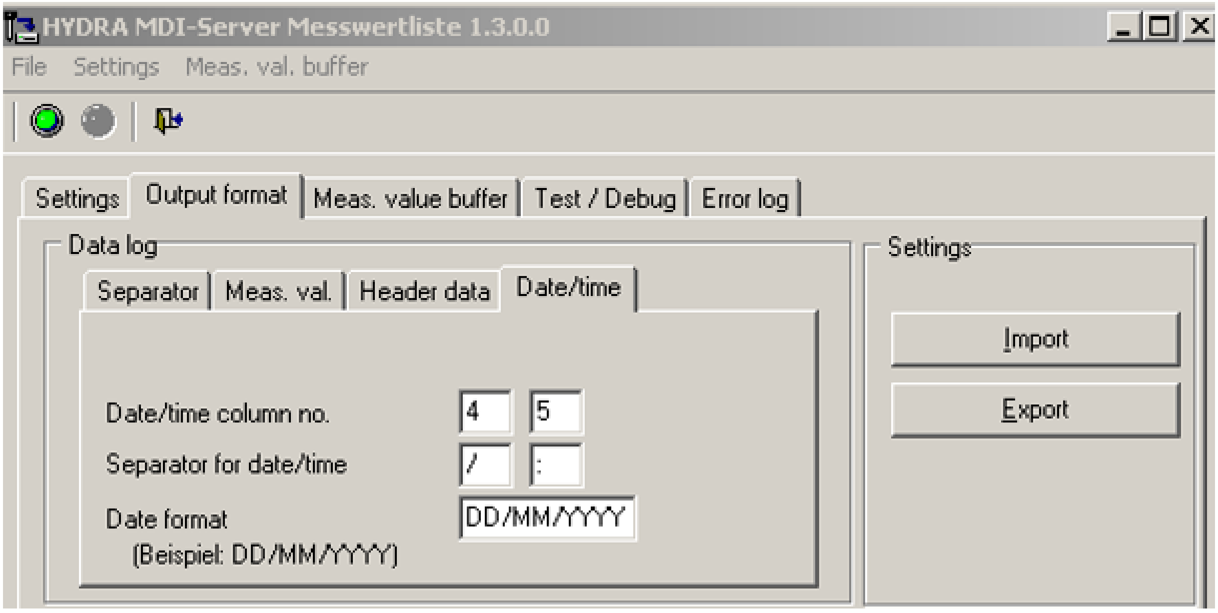
CNR	Batch number
ANR	Order number
ATK	Article number
NUM:EINTTYP	Number type for measurement recording of numbers, e. g. PPUNKT for
inspection point	
NUM:EINTNR	number, e. g. inspection point number 000001

Example for data collection via the terminal:

- The data field *CNR* is configured and the measurement buffer of a channel includes measured values with different batch numbers.

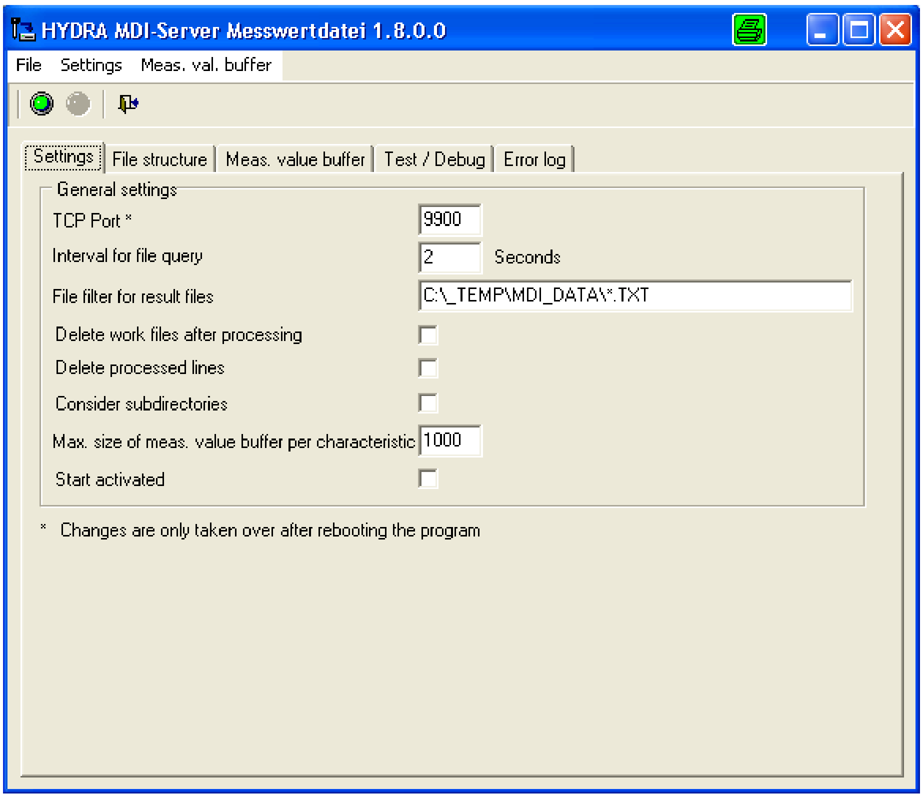
- If you open AIP inspection data collection and request measured values from the MDI server, only those measured values are collected and saved whose batch number of the MDI measurement buffer is identical to that of the inspection requirement.

- If the inspection requirement does not include a batch number, the filter is not enabled and all measured values are collected from the measurement buffer.



Use the *Date/time* tab to specify the column numbers and/or separators for the date, time and the date format.

1.7 MDI Messwertdatei (measurement file)



This MDI has been designed to collect measured values from all characteristics. You should only use this MDI for this purpose. If you use the MDI for characteristics and you select the characteristic in the AIP inspection data collection, the system automatically requests the measured values, enters and saves these values in the measurement field. Please note that the configurable additional data fields are not used to filter the collected data. In this context, AIP inspection data collection always requests and collects all data.

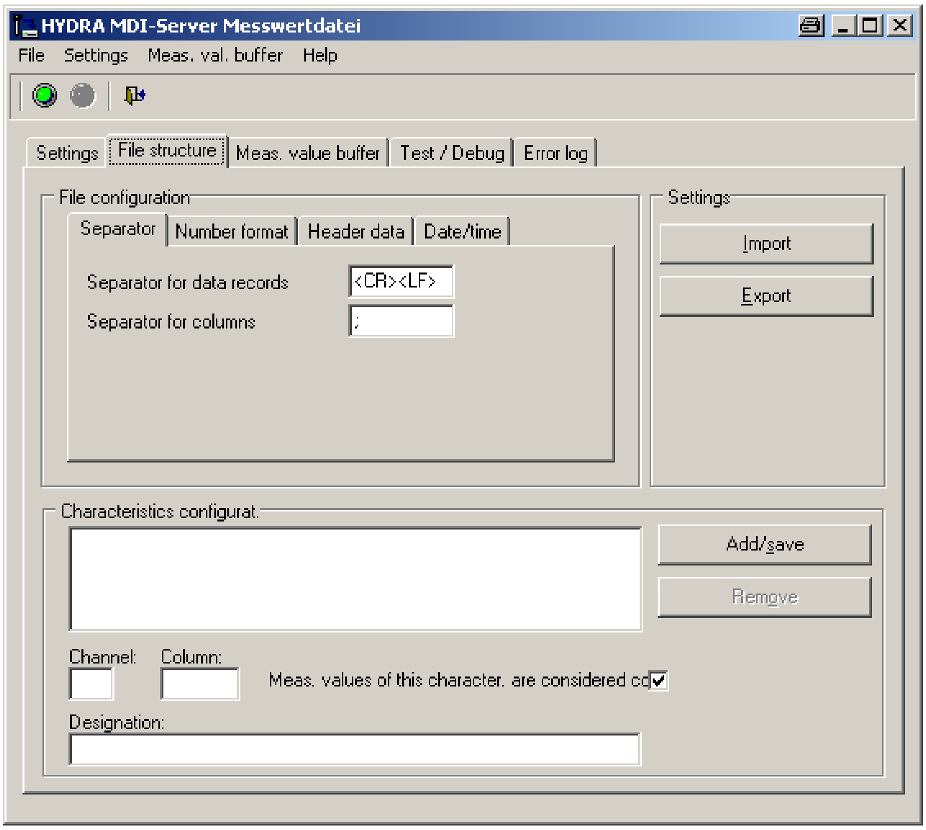
Use the *Settings* tab to define the TCP port, the interval for the file query and the file filter for result files. You can also specify if work files are deleted after processing, if subdirectories are integrated and if the MDI server is to be activated immediately upon starting. You can also specify the maximum size of the measurement buffer for each piece of measuring equipment/characteristic.

If you select the option *Delete processed lines*, data is not read from the work file but from the original file. While being accessed, the file is write-protected for other processes (locked/read-only). The read lines are deleted from the file. The first row (header) might be kept (see below).

Please note: You cannot use the options *Delete processed lines* and *Delete work files after processing* together.

Any changes to the settings are saved locally when the MDI server is closed (not when it is deactivated).

The COM port settings depend on the measuring equipment you connect. The relevant measuring equipment manuals provide further details on these settings. The entered TCP port must match the configuration of the communication port (MOC Quality Management --> Master data --> MDI Configuration --> MDI Configuration).



Use the *File structure* tab to specify the structure of a data record deriving from the measuring equipment. You can also find this information in the measuring equipment manual. Also specify how to separate the data records and/or columns of a data record.

If you want to enter special characters and/or control characters, use the synonyms listed below:

<SOH> corresponds to ASCII character 1

<STX> corresponds to ASCII character 2

<ETX> corresponds to ASCII character 3

<EOT> corresponds to ASCII character 4

<ACK> corresponds to ASCII character 6

<TAB> corresponds to ASCII character 9

<LF> corresponds to ASCII character 10

<FF> corresponds to ASCII character 12

<CR> corresponds to ASCII character 13

<NACK> corresponds to ASCII character 21

<EOF> corresponds to ASCII character 26

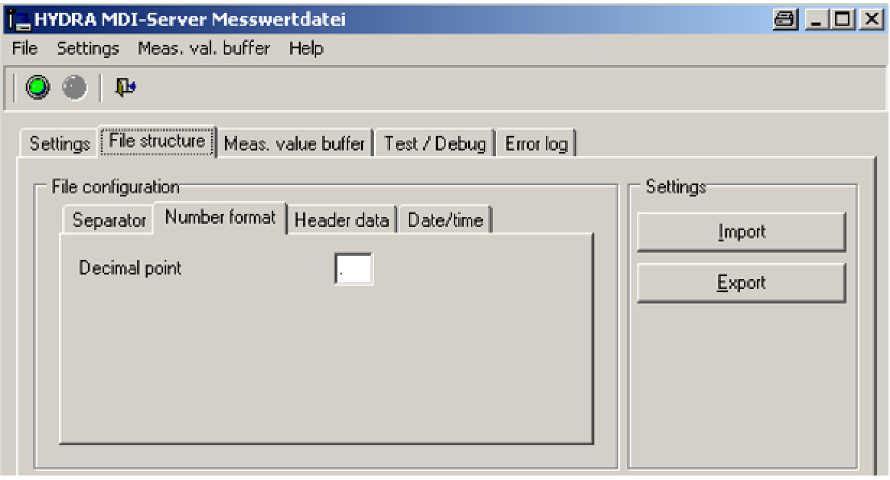
<SPACE> corresponds to ASCII character 32

Example: Enter <CR><LF> in the corresponding input field, if you want to separate the data records sent by the measuring equipment using carriage return and line feed.

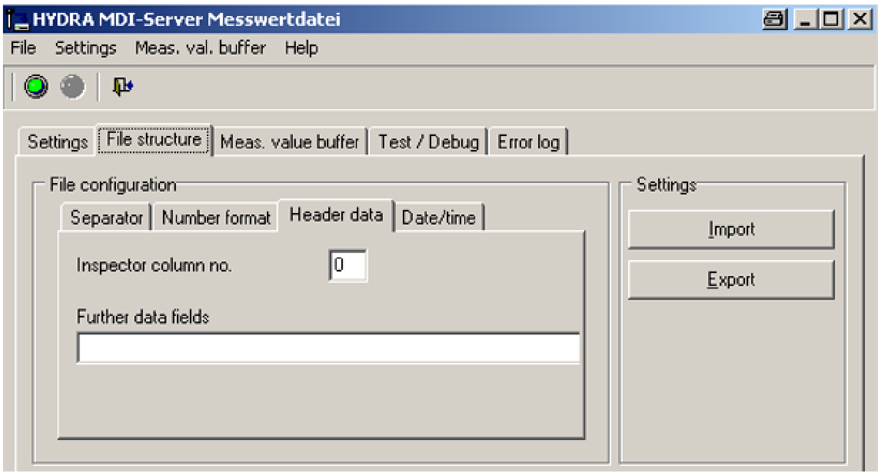
You can create and/or edit the characteristics/channels in the section *Characteristic configuration*. If you want to create a new characteristic, fill out the input fields *Channel, Column* and *Designation/Name.* If you check the corresponding option, you can specify that the measured values of the characteristic/channel are considered confirmed. Then click the *Add/save* button.

Please note: Changes to the characteristics/channels only take effect after you clicked the *Add/save* button!

The list now shows the measuring equipment (characteristic) including the channel number and the designation/name.



Use the *Number format* tab to specify the character to be used as decimal separator for the measured value. You can also find this information in the measuring equipment manual.



Enter the column number of the inspector in the *Header data* tab.

You can also specify further data fields. Use the pipe character "|" to separate these data fields. Use percent signs to indicate the column number of the data field.

Example: MNR=%2%

The machine number (MNR) is entered in the second column of the data record sent by the measuring equipment.

The terminal supports the following data fields to filter the collected data:

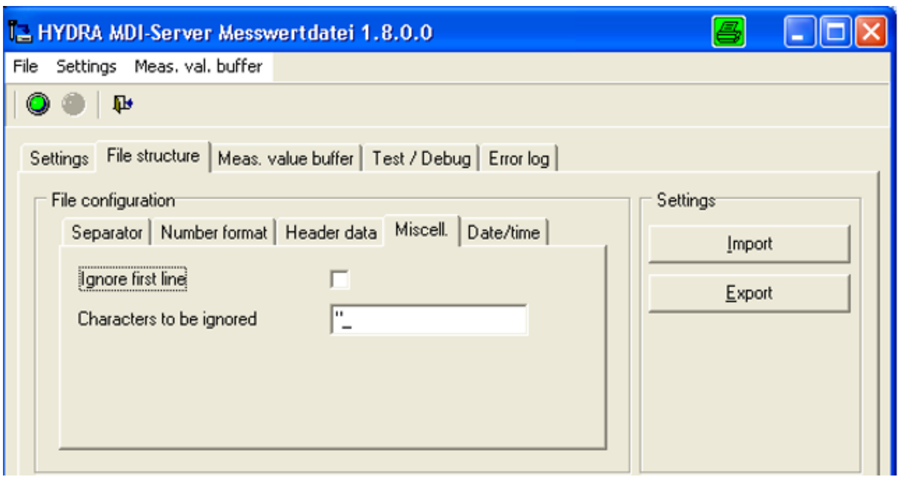
CNR	Batch number
ANR	Order number
ATK	Article number
NUM:EINTTYP	Number type for measurement recording of numbers, e. g. PPUNKT for inspection point
NUM:EINTNR	number, e. g. inspection point number 000001

Example of data collection via the terminal:

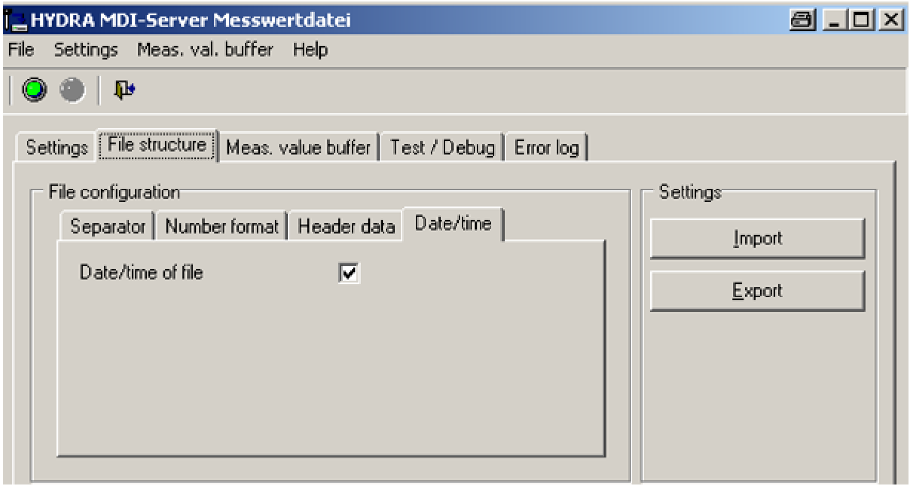
- The data field *CNR* is configured and the measurement buffer of a channel includes measured values with different batch numbers.

- If you open AIP measurement data collection and request measured values from the MDI server, only those measured values are collected and saved whose batch number of the MDI measurement buffer is identical to that of the inspection requirement.

- If the inspection requirement does not include a batch number, the filter is not enabled and all measured values are collected from the measurement buffer.



Use the *Miscell.* tab to specify whether or not the first line is to be ignored. The first line remains, if you edit the original file (see settings)! Enter the characters that should be ignored in the single data records in the field *Characters to be ignored.* You can directly enter the characters without using separators.



Use the *Date/time* tab to specify if the date and/or time of measurement should be used from the measurement data record.

2 MOC Configuration

The following example illustrates how the MDI driver measurement list is configured. The displayed configuration shows how data deriving from a measuring machine is transferred to the AIP input dialog. The displayed settings are not fixed and can be adjusted according to current requirements. You can also use a specific piece of test equipment instead of a group. You can change the AIP buttons and the driver does not have to be installed on local host IP:127.0.0.1, etc.

	Nominal value		ISO 286	Upper tolerance	Lower tolerance	Actual value	Deviation	%Deviation
Length	x	10		0.2	-0.2	10,02	0	0%
Width	y	150		0.2	-0.2	150,01	0	0%
Height	x	45		0.2	-0.2	44,98	-0.001	0%
Diameter 12	y	12		0.2	-0.2	12,02	0.016	8%
Diameter 72	x	72		0.2	-0.2	72,554	-0.02	-10%
Ledge 45	y	45		0.2	-0.2	44,997	0.001	1%
Length 123	x	123		0.2	-0.2	123,1	-0.002	-1%
Flatness	y	0.002		0.2	-0.2	-0,00225	-0.056	-28%

Figure 1: Example of a measurement list

2.1 Test equipment (gage) group

Create a test equipment group for each characteristic you want to transfer.

MOC --> Quality management -->Test equipment management --> Test equipment (gage) group

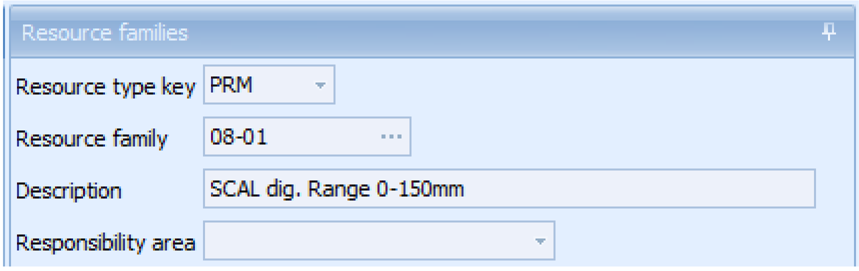


Figure 2: Example of test equipment groups

2.2 Inspection plan

Assign a measuring/test equipment group to each characteristic in the inspection plan.

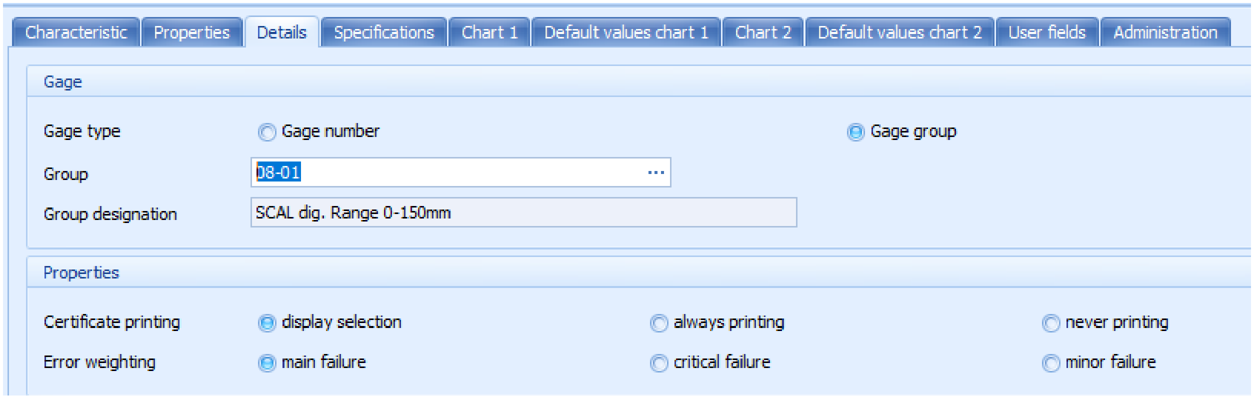


Figure 3: Example of assigning a test equipment group to a characteristic.

2.3 MDI configuration

Assign an MDI channel for each characteristic.

MOC --> Master data --> Quality management --> MDI configuration

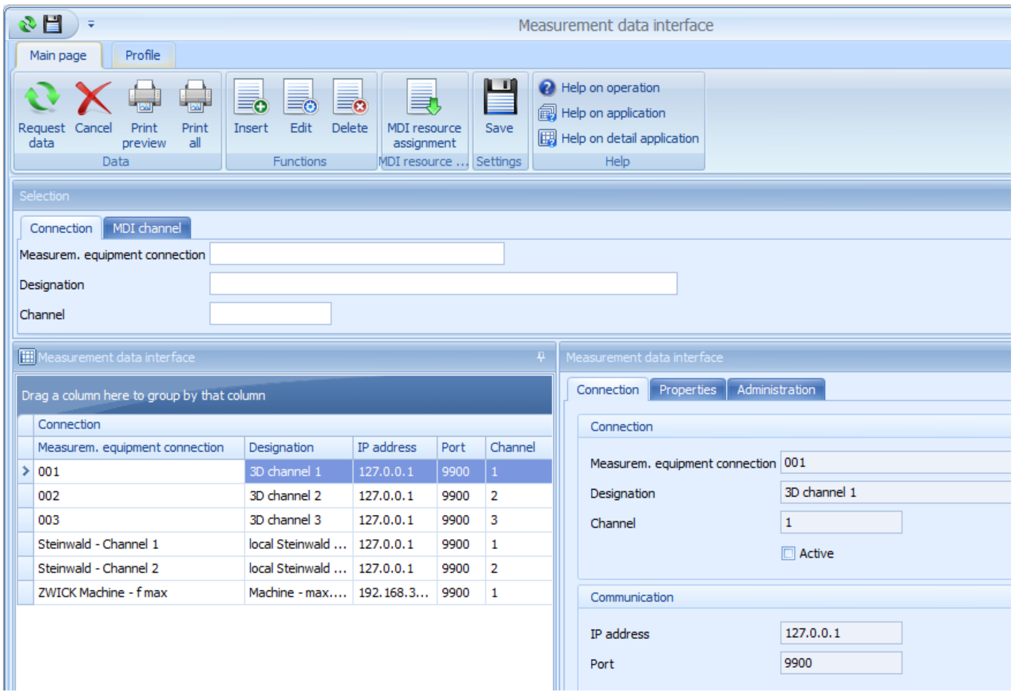


Figure 4: MDI configuration

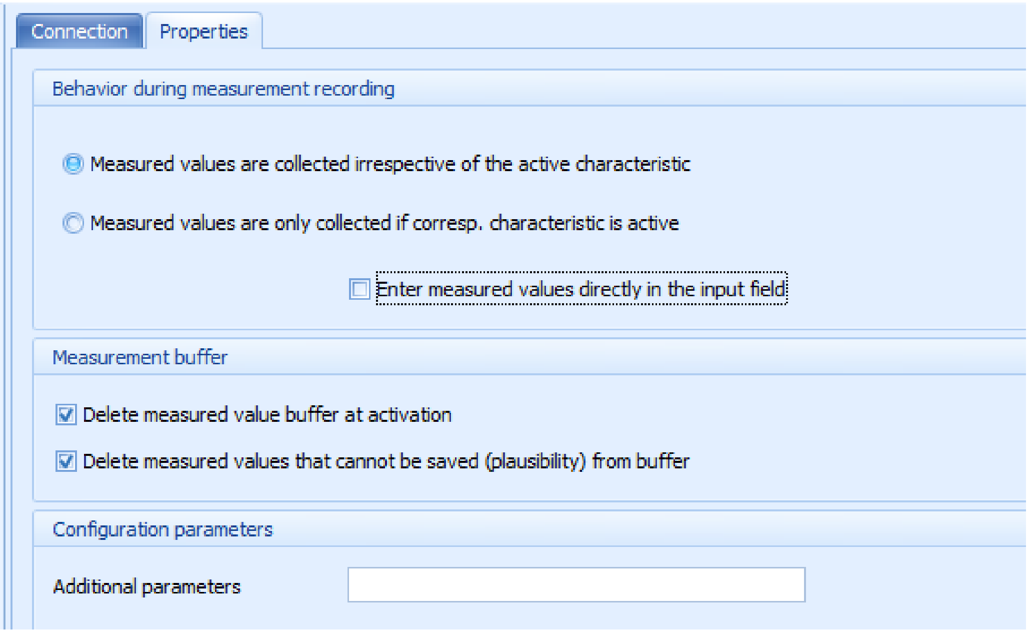


Figure 5: MDI configuration – Properties

Assign a test equipment group to each configured channel.

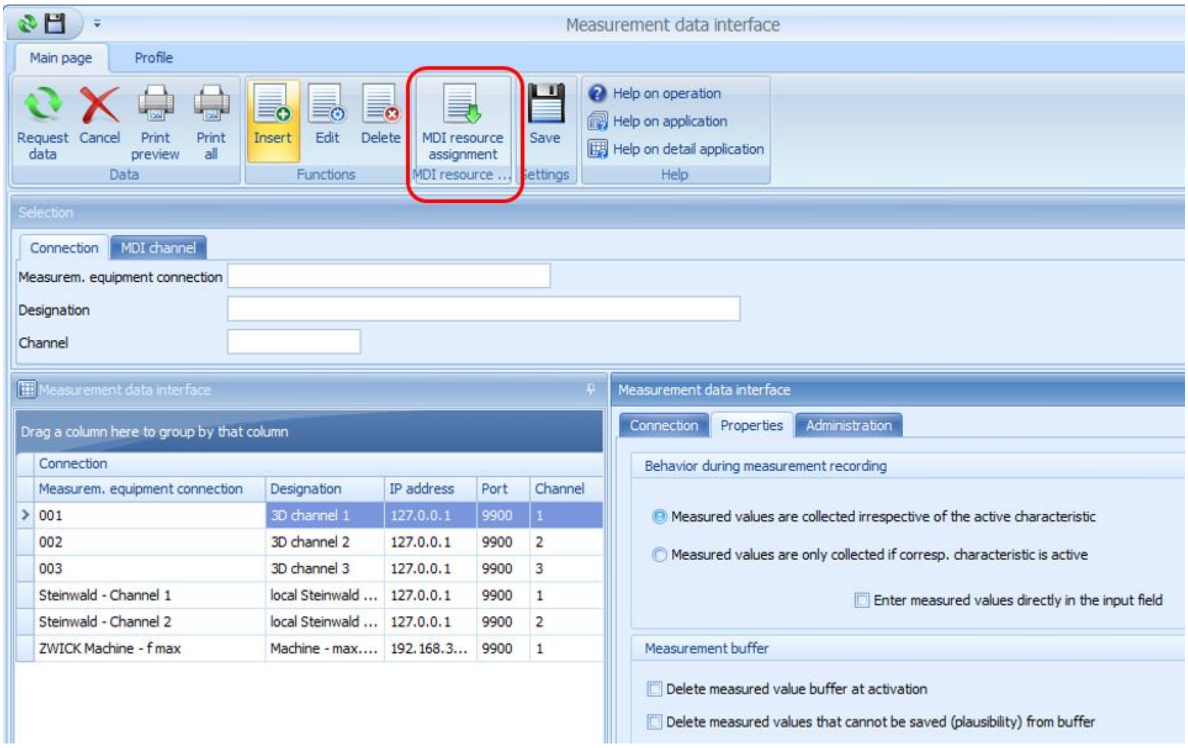


Figure 6: MDI configuration – MDI resource assignment

Use the *MDI resource assignment* application to assign pieces of test equipment or test equipment groups to the MDI channel. "Add" and/or select the test equipment group (or the piece of test equipment) to assign it to the MDI channel.

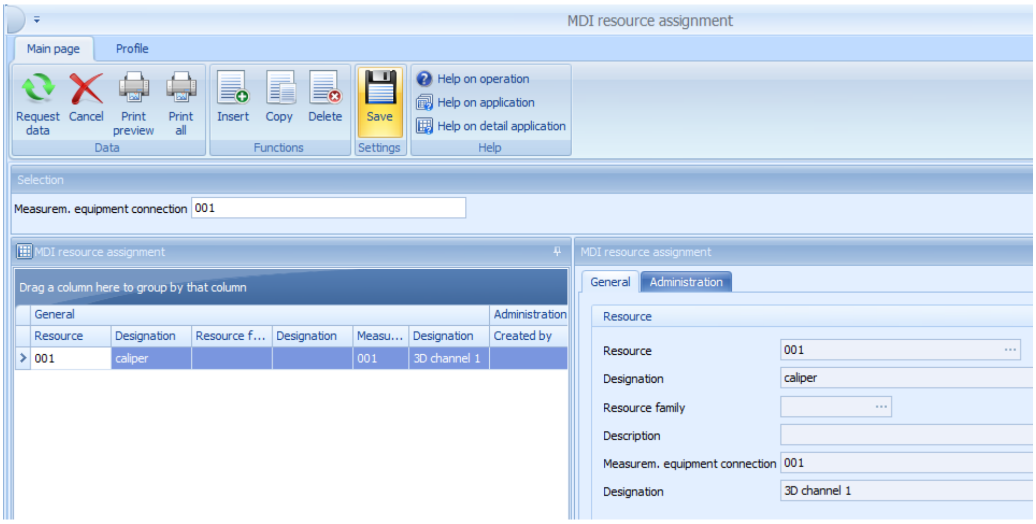


Figure 7: MDI resource assignment

3 AIP measurement data transfer

Go to the inspection point in the inspection list (left-hand side) to transfer the measured values to the AIP.

Go to the second tab of the inspection point to click the button *Accept measurement data.* Click this button to store the data from the MDI driver in the database. Click the button *Update display* to show the values.

4 MDI queue mode

The MDI queue mode is available as of service pack 14.

If you activate the MDI queue mode, the collection of measured values might be delayed as MDI measurement data are processed sequentially.

Example:
Measurement data transfer to MDI "X“ is started at AIP inspection station "A“. This process takes approx. 2 minutes.

10 seconds after starting the measurement transfer from AIP inspection station "A", AIP inspection station "B" also starts a measurement transfer to MDI "X". This transfer also takes approx. 2 minutes.

AIP inspection station "B" starts collecting the measured values 1 minute and 50 seconds after triggering the measurement transfer due to the sequential processing. The collection of measured values for AIP inspection station "B" only starts once all measured values of AIP inspection station "A" have been collected. Therefore, AIP inspection station "B" only shows all collected measured values after approx. 3 minutes and 50 seconds.

In some rare cases, problems might occur if you collect data at the same time.

4.1 Activating the MDI queue mode

4.1.1 Execute the patch "dbp_caq_mdi_lrv_queue.hsc“

Execute the database patch "dbp_caq_mdi_lrv_queue.hsc“ as follows:

Windows: cd %HYDRADIR%

hydscr.exe .\db_sql\ dbp_caq_mdi_lrv_queue.hsc > dbp_caq_mdi_lrv_queue.pro

Linux cd $HYDRADIR

hydscr.out .\db_sql\ dbp_caq_mdi_lrv_queue.hsc > dbp_caq_mdi_lrv_queue.pro

4.1.2 Changing the MDI server script

Change the MDI server script "hy_cmdilrv_auft.scr“ as follows in the HYDRA system directory.

The following screenshots indicates the part of the script that has to be changed.

```
…
HY_PATH=${HYDRADIR:=.}
FULLERRPATH=${HYFULLERRPATH:=$HY_PATH/err}

$HY_PATH/hy_cmdilrv.exe $1 $2 $3 $4 $5 $6 $7 $8 &
# $HY_PATH/hy_cmdilrv.exe -d -DD+t $1 $2 $3 $4 $5 $6 $7 $8 > $FULLERRPATH/cmdilrv_auftrag_protokoll.pro &
```

Change and/or add the paragraphs highlighted in yellow in the following screenshot.

```
…
HY_PATH=${HYDRADIR:=.}
FULLERRPATH=${HYFULLERRPATH:=$HY_PATH/err}

cmdpartmp=$1
cmdpartmp=${cmdpartmp%?}
cmdpartmp+="|TCP_COMM_REP=50|MODUS_QUEUE=1|DETAILED_LOGGING=1"

# $HY_PATH/hy_cmdilrv.exe $1 $2 $3 $4 $5 $6 $7 $8 &
# $HY_PATH/hy_cmdilrv.exe -d -DD+t $1 $2 $3 $4 $5 $6 $7 $8 > $FULLERRPATH/cmdilrv_auftrag_protokoll.pro &
$HY_PATH/hy_cmdilrv.exe $cmdpartmp $2 $3 $4 $5 $6 $7 $8 &
# $HY_PATH/hy_cmdilrv.exe -d -DD+t $cmdpartmp $2 $3 $4 $5 $6 $7 $8 > $FULLERRPATH/cmdilrv_auftrag_protokoll.pro &
```

4.2 Description of the function

How to collect measured values in the MDI queue mode:

1. The system triggers the measurement transfer via the AIP
2. The server identifies if an MDI server process is already running.
3. The system enters the request for collecting measured values in the table *caq_mdi_lrv_queue* with status *open.*

4. If no MDI server process (response to issue 2 = no) is running, the system immediately processes the new, outstanding entry of the table *caq_mdi_lrv_queue.* After processing, the system checks if the table *caq_mdi_lrv_queue* includes new entries with status open. If there are further entries with the status open, they will be processed one after the other. The MDI server process is finished automatically if the table *caq_mdi_lrv_queue* does not include an entry with the status open.
5. If an MDI server process is already running (response to issue 2 = yes), the additionally started server process will be finished automatically.

The following data is stored in the table *caq_mdi_lrv_queue* every time measured values are requested to be collected:

rec_type = area type of the inspection requirement, e. g. "FEP“

bereich = area of the inspection requirement, e. g. "F“

pruefanf_nr = unique inspection requirement number

eintrag_nr = unique inspection point number

verarb_status = processing status

anforderung_usr = HYDRA user. The HYDRA user indicates the requesting terminal number.

anforderung_ts = point in time (date and time in hh:mm:ss,123) of the request to collect measured values

verarb_start_ts = start time (date and time in hh:mm:ss,123) of processing

verarb_end_ts = end time (date and time in hh:mm:ss,123) of processing

anz_mw_success = number of measured values collected successfully

prot_filename = link to the corresponding log file

first_problem = excerpt of the log file about the first problem occurred